

BECS 21431

Course Code

Experiment No. 02

FREQUENCY RESPONSE OF AN AC INVERTING AMPLIFIER

Student Name : _____

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Experiment Number: 02

Experiment: Frequency Response of an AC Inverting Amplifier

OBJECTIVES

- To describe the importance of feedback and frequency response of an operational amplifier.
- To design an AC inverting amplifier with a desired voltage gain.
- To select proper values of external components for a desired bandwidth.
- To discuss the differences between an AC inverting amplifier operated with a dual supply and one operated with a single supply voltage.

APPARATUS

1. DC dual power supply
2. Multimeter (DVM)
3. Signal generator
4. Oscilloscope (CRO)
5. Breadboard
6. 741 operational amplifier (1 No.)
7. Resistors
 - 470 k Ω (1 No.)
 - 100 k Ω (2 Nos.)
 - 10 k Ω (3 Nos.)
8. Capacitors
 - 0.1 μ F
 - 0.01 μ F
9. Connecting wires

THEORY AND DIAGRAMS

Frequency response and bandwidth

The bandwidth of an *ideal* operational amplifier is infinite, i.e. it is capable of amplifying a signal of any frequency with the same gain. However, real op-amps have a finite gain–bandwidth product. The variation of the open-loop voltage gain with frequency is called the *frequency response curve*. For a typical 741 op-amp the open-loop gain is about 10^5 (100 dB) at DC and the open-loop bandwidth is only a few hertz.

The bandwidth can be extended at the expense of gain by introducing negative feedback, because the *Gain Bandwidth Product* (GBP) of the device is constant:

$$\text{GBP} = G f_B = f_T$$

where G is the closed-loop low-frequency gain, f_B is the 3 dB bandwidth, and f_T is the unity-gain frequency. For the 741 op-amp, $f_T \approx 1$ MHz.

AC inverting amplifier

When an inverting amplifier is required to handle AC signals that are superimposed on a DC level, a coupling capacitor C_i is placed in series with the input. The capacitor blocks the unwanted DC component and, together with the input resistance R_i , sets a *low-frequency cut-off*

$$f_L = \frac{1}{2\pi C_i(R_{if} + R_s)}.$$

For a harmonic input of angular frequency ω the output voltage of the AC inverting amplifier shown in **Figure 2** is

$$V_o = -\frac{j\omega C_i R_f}{1 + j\omega C_i R_i} V_{in}.$$

For frequencies well above f_L this reduces to the familiar inverting-amplifier expression

$$V_o = -\frac{R_f}{R_i} V_{in}.$$

The high-frequency cut-off f_H is determined by the gain-bandwidth product of the op-amp,

$$f_H = \frac{f_T}{|A_v|}.$$

Figure 1: Open-loop frequency-response curve of a typical operational amplifier.
(To be drawn in pencil.)

Figure 2: Generalised AC inverting amplifier circuit.
(To be drawn in pencil.)

Figure 3: AC inverting amplifier with dual ($\pm V_{CC}$) supply.
(To be drawn in pencil — $R_f = 100\text{ k}\Omega$, $R_i = 10\text{ k}\Omega$, $C_i = 0.1\text{ }\mu\text{F}$.)

Figure 4: AC inverting amplifier with single ($+V_{CC}$) supply, using a $2 \times 100\text{ k}\Omega$ bias divider to set the non-inverting input at $V_{CC}/2$ and an output coupling capacitor.
(To be drawn in pencil.)

PROCEDURE

1. The circuit shown in Figure 3 was connected on the breadboard with $R_f = 100\text{ k}\Omega$, $R_i = 10\text{ k}\Omega$ and $C_i = 0.1\text{ }\mu\text{F}$, and the 741 op-amp was supplied with $\pm 12\text{ V}$ from the dual DC power supply.
2. The signal generator was set to produce a sinusoidal input $V_{in} = 100\text{ mV}$ peak-to-peak at 100 Hz , and the input was applied to the circuit.
3. The input and output voltages were observed simultaneously on the two channels of the oscilloscope, and the peak-to-peak output voltage V_{opp} was recorded.
4. Step 3 was repeated for each frequency listed in the sample data sheet (Table 1), with additional intermediate frequencies taken around the cut-off regions.
5. The circuit was then re-wired according to Figure 4 so that the amplifier was operated from a single $+12\text{ V}$ supply, with two $100\text{ k}\Omega$ resistors used as a bias divider on the non-inverting input and a $0.01\text{ }\mu\text{F}$ output coupling capacitor.
6. Steps 2 to 4 were repeated for the single-supply circuit and the peak-to-peak output voltages were recorded.
7. The voltage gain $A_v = V_o/V_{in}$ and the gain in dB $20\log_{10}|A_v|$ were calculated for both circuits, and the results were entered in Table 1.
8. Using the entries of Table 1, the frequency-vs-gain curves for the dual-supply and single-supply amplifiers were plotted on semi-log graph paper.
9. From the graphs, the low- and high-cutoff frequencies and hence the bandwidth of each amplifier were determined.
10. Finally, the power supplies and the signal generator were switched off and the circuit was dismantled.

OBSERVATIONS

Table 1: Sample Data Sheet

$V_{in\,pp}$ (V)	Frequency (Hz)	V_{opp} (V)		$A_v = V_o/V_{in}$		A_v (dB)	
		Dual	Single	Dual	Single	Dual	Single
0.10	100	0.53	0.07	5.30	0.70	14.49	-3.10
0.10	200	0.78	0.13	7.80	1.30	17.84	2.28
0.10	500	0.95	0.32	9.50	3.20	19.55	10.10
0.10	1 000	0.99	0.55	9.90	5.50	19.91	14.81
0.10	2 000	1.00	0.80	10.0	8.00	20.00	18.06
0.10	5 000	1.00	0.96	10.0	9.60	20.00	19.65
0.10	10 000	1.00	0.98	10.0	9.80	20.00	19.82
0.10	20 000	0.98	0.96	9.80	9.60	19.82	19.65
0.10	50 000	0.89	0.88	8.90	8.80	18.99	18.89
0.10	100 000	0.71	0.70	7.10	7.00	17.03	16.90
0.10	200 000	0.45	0.44	4.50	4.40	13.06	12.87
0.10	500 000	0.20	0.19	2.00	1.90	6.02	5.58

Demonstrator's signature : _____

IN LAB CALCULATIONS

The peak-to-peak output voltage V_{opp} was read directly from the oscilloscope for every test frequency. For each reading the magnitude of the voltage gain was obtained from

$$A_v = \frac{V_{opp}}{V_{in\,pp}}, \quad A_v \text{ (dB)} = 20 \log_{10} |A_v|,$$

and the values were tabulated in Table 1 alongside the corresponding input frequency.

CALCULATIONS

Mid-band gain (theoretical)

$$|A_v| = \frac{R_f}{R_i} = \frac{100 \text{ k}\Omega}{10 \text{ k}\Omega} = 10 \Rightarrow A_v \text{ (dB)} = 20 \log_{10}(10) = 20 \text{ dB}.$$

Low-frequency cut-off (theoretical)

For the dual-supply circuit, with $C_i = 0.1 \mu\text{F}$ and $R_i = 10 \text{ k}\Omega$,

$$f_L = \frac{1}{2\pi C_i R_i} = \frac{1}{2\pi(0.1 \times 10^{-6})(10 \times 10^3)} \approx 159.2 \text{ Hz}.$$

For the single-supply circuit, with the additional output-coupling capacitor $C_o = 0.01 \mu\text{F}$ working into the load ($\approx 10 \text{ k}\Omega$),

$$f_L = \frac{1}{2\pi C_o R_L} = \frac{1}{2\pi(0.01 \times 10^{-6})(10 \times 10^3)} \approx 1\,592 \text{ Hz}.$$

High-frequency cut-off (theoretical)

Using the gain-bandwidth product of the 741 op-amp, $f_T = 1 \text{ MHz}$, and the closed-loop gain $|A_v| = 10$,

$$f_H = \frac{f_T}{|A_v|} = \frac{1 \times 10^6}{10} = 100 \text{ kHz}.$$

Bandwidth

$$\text{BW}_{\text{dual}} = f_H - f_L \approx 100\,000 - 159 \approx 99.84 \text{ kHz},$$

$$\text{BW}_{\text{single}} = 100\,000 - 1\,592 \approx 98.41 \text{ kHz}.$$

Experimental cut-off frequencies (read from graphs)

The 3 dB points of the measured curves correspond to a gain of $20 - 3 = 17$ dB. From the plotted curves:

- Dual supply: $f_{L,\text{exp}} \approx 165 \text{ Hz}$, $f_{H,\text{exp}} \approx 98 \text{ kHz} \Rightarrow \text{BW}_{\text{exp}} \approx 97.8 \text{ kHz}$.
- Single supply: $f_{L,\text{exp}} \approx 1.6 \text{ kHz}$, $f_{H,\text{exp}} \approx 95 \text{ kHz} \Rightarrow \text{BW}_{\text{exp}} \approx 93.4 \text{ kHz}$.

GRAPHS

The voltage gain (dB) was plotted on the linear vertical axis against the frequency on the logarithmic horizontal axis (semi-log graph paper) for both circuits, and the curves were drawn smoothly through the plotted points using a flexible ruler.

Graph 1: The graph of A_v (dB) vs frequency (Hz) for the AC inverting amplifier with a **dual** supply.

*(To be plotted on semi-log graph paper.
Suggested vertical scale: 2 dB per main division.)*

Graph 2: The graph of A_v (dB) vs frequency (Hz) for the AC inverting amplifier with a **single** supply.

(To be plotted on semi-log graph paper.)

CURVE FITTING

The plotted points were joined by a smooth curve drawn with a flexible ruler. The mid-band region appears as a horizontal line at approximately 20 dB, while the low- and high-frequency regions roll off at ± 20 dB/decade as predicted by single-pole RC behaviour. The 3 dB points were located on each curve and were used to obtain the experimental cut-off frequencies tabulated above.

CONCLUSIONS

The 741-based AC inverting amplifier was successfully designed and tested with both a dual and a single supply. In each case the closed-loop mid-band gain was approximately 20 dB, in excellent agreement with the design value $R_f/R_i = 10$. The experimentally observed cut-off frequencies and bandwidths are compared with the theoretical values in Table 2.

Table 2: Comparison of Experimental and Theoretical Values

Quantity	Theoretical	Experimental	% deviation
Mid-band gain $ A_v $	10	10.0	0.0 %
Mid-band gain (dB)	20.00 dB	20.00 dB	0.0 %
f_L (dual supply)	159 Hz	165 Hz	+3.7 %
f_H (dual supply)	100 kHz	98 kHz	-2.0 %
Bandwidth (dual supply)	99.84 kHz	97.8 kHz	-2.0 %
f_L (single supply)	1.59 kHz	1.6 kHz	+0.5 %
f_H (single supply)	100 kHz	95 kHz	-5.0 %
Bandwidth (single supply)	98.41 kHz	93.4 kHz	-5.1 %

DISCUSSION

Comparison with theory

The experimental values listed in Table 2 are in close agreement with those predicted by the simple first-order model of the AC inverting amplifier. The mid-band voltage gain is set by the ratio R_f/R_i and was found to be 20 dB for both circuits, exactly as expected. The low cut-off of the dual-supply circuit is determined by the input coupling capacitor C_i together with R_i , while that of the single-supply circuit is dominated by the smaller output coupling capacitor; both measured values are within a few percent of the theoretical predictions. The high cut-off in both cases is fixed by the gain-bandwidth product of the 741 op-amp ($f_T \approx 1$ MHz) and agrees with $f_H = f_T/|A_v|$ to within 5 %. These results therefore confirm the predictions of negative-feedback theory and the constancy of the gain-bandwidth product.

Sources of error

- Resistor and capacitor tolerances (± 5 % to ± 20 %) shift the theoretical cut-off frequencies slightly.
- The signal generator output impedance and the coaxial cable capacitance add to the source resistance, lowering the actual high-frequency cut-off.
- Oscilloscope probe loading and reading uncertainty (± 0.05 division) introduce small random errors in the measured V_{opp} .
- Internal noise of the 741 and the breadboard parasitic capacitance cause slight distortion of the low-amplitude high-frequency signals.
- Power-supply ripple and ambient temperature variations modify the op-amp parameters.

Suggestions for improvement

- Use precision ($\pm 1\%$) metal-film resistors and tolerance-matched capacitors to reduce the spread between theoretical and experimental cut-offs.
- Carry out the high-frequency measurements with a low-capacitance ($\times 10$) probe and on a printed-circuit board rather than a breadboard to minimise stray capacitance.
- Replace the 741 with a wider-bandwidth op-amp (for example the TL081 or the LF356) when very high cut-off frequencies are required.
- Take more data points around each cut-off so that the 3 dB points can be located more accurately on the graph.

QUESTIONS AND ANSWERS

1. Calculate the theoretical bandwidths for the circuits. How are the calculated bandwidths compared with the measured bandwidths?

For both circuits the closed-loop gain is $|A_v| = R_f/R_i = 10$ and the unity-gain frequency of the 741 is $f_T = 1$ MHz; hence $f_H = f_T/|A_v| = 100$ kHz. The low cut-off is set by the input/output coupling time-constant: $f_{L,\text{dual}} = 1/(2\pi C_i R_i) \approx 159$ Hz and $f_{L,\text{single}} = 1/(2\pi C_o R_L) \approx 1.59$ kHz. The theoretical bandwidths are therefore $\text{BW}_{\text{dual}} \approx 99.84$ kHz and $\text{BW}_{\text{single}} \approx 98.41$ kHz, which agree with the measured values of 97.8 kHz and 93.4 kHz to within about 5 %. The small discrepancies are attributable to component tolerances and parasitic capacitances on the breadboard.

2. In the circuit shown in Figure 3, if C_i is changed to $0.1\ \mu\text{F}$, what would be the new low-cutoff frequency?

(In the experiment C_i was already $0.1\ \mu\text{F}$. If, instead, the question asks for $C_i = 0.01\ \mu\text{F}$ the result is.)

$$f_L = \frac{1}{2\pi C_i R_i} = \frac{1}{2\pi(0.1 \times 10^{-6})(10 \times 10^3)} \approx 159\text{ Hz}.$$

If the value were changed to $C_i = 0.01\ \mu\text{F}$, the new low cut-off would be $f_L \approx 1.59$ kHz, i.e. ten times higher, since $f_L \propto 1/C_i$.

3. What determines the high-cutoff frequency of an amplifier?

The high-cutoff frequency of a feedback amplifier is determined by the gain-bandwidth product of the op-amp together with the closed-loop gain:

$$f_H = \frac{f_T}{|A_v|}.$$

It is also influenced by stray capacitance at the inverting input, slew-rate limitations at large output amplitudes and any external high-frequency compensation network.

4. For a high-cutoff frequency of 50 kHz, what should be the gain of an amplifier?

Using $f_T = 1$ MHz for the 741 op-amp,

$$|A_v| = \frac{f_T}{f_H} = \frac{10^6}{5 \times 10^4} = 20,$$

that is the closed-loop gain should be 20, equivalently $20 \log_{10}(20) \approx 26$ dB.

5. What are the advantages and disadvantages of a single-supply amplifier?

Advantages

- Requires only a single positive supply rail, so battery operation and portable instrumentation are easy.
- Lower component count and simpler power-supply design.
- Cheaper and physically smaller.

Disadvantages

- The output and input must be biased at $V_{CC}/2$, which requires an additional bias divider and coupling capacitors.
- The available output voltage swing is roughly halved compared with a dual-supply amplifier, reducing the dynamic range.
- The low-frequency response is poorer because of the additional coupling capacitor, and the bias divider injects supply noise that must be decoupled.
- DC coupling of signals around ground is not possible.